

MongoDB Sharded Cluster Engineering Guide

Executive Overview, Architecture Governance & Readiness Roadmap

AUDIENCE TARGET: TEAM LEADER / ENGINEERING MANAGER

1. Executive Architecture Summary (14-Node DC Topology)

This operational roadmap details the architecture, high-availability properties, and governance strategy for our production MongoDB Community Edition sharded cluster across Primary and Disaster Recovery (DR) Data Centers.

Cluster Component	Primary DC Nodes	DR DC Nodes	HA / Quorum Rule
Config Servers (CSRS)	3 Nodes (csrs01-03)	1 Hidden Node (csrs04)	2/3 Majority Required
Shard 01 (Data)	3 Nodes (s1-n1,n2,n3)	1 Hidden Node (s1-dr)	Tolerates 1 Node Loss
Shard 02 (Data)	3 Nodes (s2-n1,n2,n3)	1 Hidden Node (s2-dr)	Tolerates 1 Node Loss
Shard 03 (Data)	3 Nodes (s3-n1,n2,n3)	1 Hidden Node (s3-dr)	Tolerates 1 Node Loss
mongos Routers	2 Active Routers	2 Standby Routers	HAProxy VIP Balancing

2. Business Continuity, Quorum Math & Disaster Recovery

Replica Set Quorum Math: Each of the 3 shards has 3 voting nodes in the Primary DC. Majority quorum is 2 out of 3. If a single primary node crashes, an automated election completes in 3–5 seconds with 0 data loss.

CSRS Quorum Hazard: If 2 out of 3 Config Server nodes crash, cluster metadata locks into *Read-Only* mode. Application reads continue, but collection creation, chunk splits, and balancing pause until emergency force reconfig.

DC-DR Strategy: Stretched cluster model with priority 0 hidden nodes in DR, complemented by **Percona Backup for MongoDB (PBM)** for continuous physical backups and Point-In-Time Recovery (PITR).

3. Team Engineering Readiness & Enablement Roadmap

To ensure 24/7 operational readiness, the engineering team must complete a 4-stage enablement program:

Stage	Program Phase	Target Outcome / Deliverable
Phase 1	Local Docker Sandbox Setup	Deploy complete 14-node cluster locally using docker-compose.yml
Phase 2	Chaos Engineering Drills	Inject primary crashes, CSRS drops, oplog bursts via chaos tool
Phase 3	Ticket RCA Protocol	Submit standardized incident RCA reports (rca_template.md)

Phase 4	Production SRE Audit	Execute automated audit script (mongodb_cluster_audit.py) monthly
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